Does DVOL add information beyond HAR?

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Abstract. Raw Bitcoin (BTC) DVOL, Deribit's option-implied volatility index, has the lowest mean quasi-likelihood (QLIKE) variance loss in this sample: 0.09611, compared with 0.11123 for the fitted augmentation. The comparison uses 1,357 matched daily forecasts from 2023 to 2026 and a direct 30-day heterogeneous autoregressive (HAR) benchmark built from price-history variance. Adding DVOL lowers mean QLIKE relative to HAR by 11.35%, but the paired difference of -0.01424 has a descriptive 95% interval [-0.03490, +0.00505] that includes zero. Estimated effective sample size (ESS), which adjusts sample information for dependence, is 51.71, below the preset reporting threshold of 60. The allocation diagnostic shows no clear improvement in meeting a fixed annual volatility budget. The ESS rule excludes every declared one-year simulation from confirmatory inference.

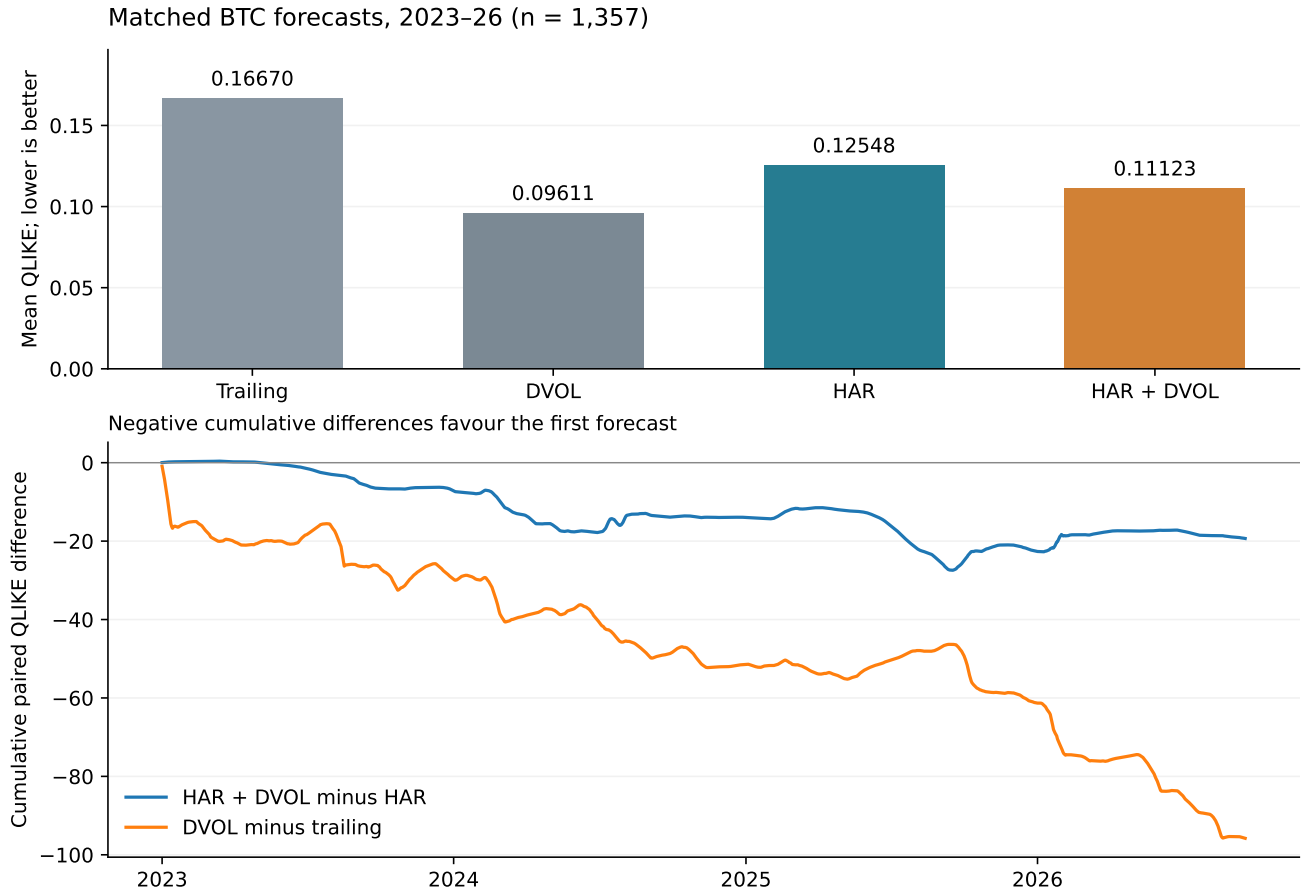


Figure 1. Mean QLIKE by forecast and cumulative paired loss differences, using the same 1,357 daily origins. Negative differences favour the first forecast named in the legend.

1 Introduction

Does option-implied variance add forecasting information beyond daily, weekly and monthly realised variance? I compare Corsi's HAR construction, adapted to a direct 30-day target, with the same regression augmented by the previous completed daily DVOL variance. Raw DVOL and trailing variance provide reference forecasts.

The most striking point-loss ranking is raw DVOL ahead of both fitted regressions. Augmentation reduces mean QLIKE relative to HAR, while its uncertainty interval spans zero and the allocation diagnostic shows a higher point estimate of risk-budget error. The lower loss of raw DVOL highlights estimation risk in the fitted augmentation. The following sections distinguish these estimates from the limits of the evidence and then set out the locked prospective comparison.

2 Data and method

The retained inputs contain 48,626 hourly BTC-PERPETUAL candles and 1,996 daily DVOL candles, downloaded on 19 October 2026. Evaluation uses daily midnight forecast origins from 1 January 2023 through 18 September 2026. Each target is the next 720 hourly log-return squares, annualised by 8,760; that window requires 721 completed closes.

Both regressions use exactly the latest 365 labels whose full 30-day targets have ended. Prior 1-, 7- and 30-day annualised variances form the HAR features. The augmented model adds the previous completed daily DVOL close divided by 100 and then squared. Level ordinary least squares includes an intercept, with features centred and scaled using training rows only. An annual-variance floor of 1e-6 applies to forecasts; raw predictions and every floor hit are retained. Neither regression reaches that floor in this sample.

With r_t denoting an hourly log return and t the forecast origin, the future target and completed-return features are

$$V_t^{(30\mathrm{d})} = \frac{8760}{720}\sum_{h=1}^{720} r_{t+h}^2,$$

$$RV_t^{(k)} = \frac{8760}{24\,k}\sum_{h=1}^{24\,k} r_{t-h+1}^2, \quad k \in \{1, 7, 30\}.$$

The two direct forecasts are

$$\widehat{V}_t^{(30\mathrm{d}),\mathrm{HAR}} = \widehat{\beta}_{0,t} + \widehat{\beta}_{d,t} RV_t^{(1)} + \widehat{\beta}_{w,t} RV_t^{(7)} + \widehat{\beta}_{m,t} RV_t^{(30)},$$

$$\widehat{V}_t^{(30\mathrm{d}),\mathrm{HAR+DVOL}} = \widehat{\gamma}_{0,t} + \widehat{\gamma}_{d,t} RV_t^{(1)} + \widehat{\gamma}_{w,t} RV_t^{(7)} + \widehat{\gamma}_{m,t} RV_t^{(30)} + \widehat{\gamma}_{D,t}\left(\frac{\mathrm{DVOL}_t^{\mathrm{prev.close}}}{100}\right)^2.$$

All comparisons use identical origins. Smaller values of the QLIKE loss indicate more accurate variance forecasts:

$$\mathrm{QLIKE} = \frac{\text{realised}}{\text{forecast}} - \log\left(\frac{\text{realised}}{\text{forecast}}\right) - 1.$$

3 Raw DVOL has the lowest average loss

Table 1 reports the matched forecast comparison. Augmentation minus HAR has mean -0.01424, with a 95% interval [-0.03490, +0.00505]. Raw DVOL has the lowest point loss, including against the fitted augmentation.

Table 1. Matched variance forecasts. Lower mean QLIKE indicates better forecast accuracy.

Forecast	Mean QLIKE	Role
Trailing 30-day	0.16670	Reference
Raw DVOL	0.09611	Reference
HAR	0.12548	Benchmark
HAR + DVOL	0.11123	Augmentation

Level Mincer–Zarnowitz regressions assess forecast calibration, with heteroskedasticity- and autocorrelation-consistent (HAC) uncertainty. The diagnostics include a joint HAR calibration/encompassing restriction and

a separate incremental-coefficient restriction with the HAR slope and intercept unrestricted. Coefficients and uncertainty are retained in `summary.json`.

Large absolute differences hurt augmentation

The largest 1% and 5% of absolute augmentation loss differences account for 8.6% and 27.8% of total absolute differences. Their aggregate signed contributions are positive, so these groups hurt augmentation even though the full-sample mean is negative. This pattern runs against a simple explanation in which augmentation wins by avoiding the most extreme losses.

Table 2. Leaving out each evaluation year. The final column is mean QLIKE for HAR + DVOL minus HAR on the remaining origins.

Year left out	Remaining origins	Mean augmentation difference
2023	992	-0.01233
2024	991	-0.01260
2025	992	-0.01072
2026	1096	-0.02065

The retained diagnostics cover every fixed calendar quarter, extreme-origin dates and all 30 non-overlapping starting phases. Each phase contains 45 or 46 outcomes. These displays describe concentration; their interpretation is set out in the limitations below.

The lower forecast loss brings no clear budget gain

A diagnostic allocation rule targets an illustrative 20% annual volatility budget by scaling assumed fractional exposure. Its capped weight and squared error relative to the variance budget are

$$w = \min\left(1, \frac{0.20}{\sqrt{\text{forecast variance}}}\right), \qquad \left(w^2 \times \text{realised variance} - 0.04\right)^2.$$

Table 3. The fixed volatility-budget diagnostic. Budget exceeded is the share of origins above the target variance budget.

Forecast	Mean weight	Mean squared budget error	Budget exceeded
Trailing	0.469	0.0008186	46.7%
DVOL	0.418	0.0002872	29.6%
HAR	0.420	0.0004014	36.3%
HAR + DVOL	0.437	0.0004196	41.7%

Augmentation minus HAR has mean risk loss +0.0000182, with interval [-0.0000818, +0.0001221]. The higher point estimate and interval spanning both signs give no clear improvement under this decision loss. Entry-plus-exit fee amounts are reported separately at 0, 5, 10 and 20 basis points per side.

4 Inference limits

Paired stationary-bootstrap inference uses 5,000 draws, a fixed seed and expected blocks of 60 days, with a 59-lag HAC cross-check. Block sensitivities use 30, 90, 180 and 365 days. Overlapping targets and rolling fits can create dependence beyond the 30-day horizon. ESS estimates that dependence-adjusted information; the preset minimum-ESS rule is a prerequisite for confirmatory reporting.

Each declared synthetic scenario has 200 independent replications at sample sizes 365 and 1,357, with 2,000 bootstrap draws per replication. The population null has exactly zero mean, without centring each realised sample. The scenarios use a 30-day moving-average loss component alone, add a stationary slow component of persistence $\exp(-1/365)$, or add symmetric rare shocks spread across overlapping windows. The alternative shifts the mean by -0.25 marginal standard deviations, rather than -0.25 QLIKE.

Table 4. Synthetic null rejection rates before the ESS rule. Monte Carlo (MC) intervals describe simulation uncertainty; the last column is the share of replications passing the rule.

n	Synthetic loss process	Ungated rejection	95% MC interval	ESS rule passes
365	30-day overlap	19.0%	14.2% to 25.0%	0.0%
365	30-day overlap plus slow component	49.0%	42.2% to 55.9%	0.0%
365	30-day overlap plus clustered tail shocks	16.0%	11.6% to 21.7%	0.0%
1357	30-day overlap	10.0%	6.6% to 14.9%	1.5%
1357	30-day overlap plus slow component	45.0%	38.3% to 51.9%	0.0%
1357	30-day overlap plus clustered tail shocks	8.5%	5.4% to 13.2%	1.5%

At $n = 365$, ungated null rejection ranges from 16% to 49%, against nominal 5%. The ESS rule passes zero of 200 replications under every one-year null and alternative. The Wilson 95% upper bound is around 1.9% for the probability of passing the rule in those scenarios. At the one-year sample size, the rule prevents confirmatory reporting in these stress tests. At $n = 1,357$, some conditional estimates use only one to four passing replications, with wide Monte Carlo intervals.

5 Limitations

All historical findings and generated-forecast diagnostics are exploratory. The additional historical work followed inspection of the data, so chronological training alone leaves the evaluation exposed to research choices made after inspection. The interval estimates are descriptive, the Mincer–Zarnowitz and encompassing p-values are diagnostics rather than additional primary discoveries, and the point-loss ranking does not establish formal superiority. ESS is an estimated dependence diagnostic, not an observed count of independent episodes. The non-overlapping phase samples remain dependent; their role is to expose concentration, rather than select a favourable crisis window or identify an ex-ante regime rule. The synthetic stress tests approximate loss dependence rather than the complete rolling HAR estimation pipeline. Zero observed ESS-rule passes leave true pass probability uncertain, as the reported Wilson bound makes explicit.

The original source check compares open, high, low and close (OHLC) fields. It fails because one unused DVOL open differs; all three audited DVOL closes match, but those dates provide limited coverage of the full history. Historical downloads also leave the original unrevised vintages unverified. Returns from one perpetual contract proxy the spot index, and the study covers one asset and venue with overlapping outcomes.

Forecast accuracy and a variance-budget diagnostic define the economic scope. The latter assumes constant fractional exposure; a fixed-unit portfolio, rebalancing, funding and profit lie outside the implementation. The allocation rule has no established optimality for squared budget error. Fee amounts remain separate from that error, so the results identify neither net trading returns, a causal variance premium nor an executable edge. The cumulative figure measures loss accumulation. The separate option-distribution study retains finite calendar-grid and extrapolated-tail limitations.

6 Prospective protocol

The primary prospective contrast remains HAR + DVOL minus HAR. An inference-validation amendment disables formal superiority claims for the planned year, which collects timestamped evidence under the fixed design. A later inference design needs a separate prospective declaration; the current cohort and terminal date stay fixed even if the result is inconclusive.

The HAR cohort contains 365 scheduled origins from 21 October 2026 through 20 October 2027, each at 01:00 UTC. Features end at midnight, forecasts freeze between 00:10 and 00:50, and outcomes begin at 01:00. Training uses the correspondingly shifted historical 30-day targets completed by midnight, a different alignment from the retrospective midnight targets. Final maturity is 19 November 2027, with terminal analysis fixed at 21 November 2027, 01:00 UTC.

Each new forecast requires matching hourly and daily DVOL closes and completion of fitting and the durable write before the freeze deadline. Missed scheduled days count as missed and stay unfilled. Terminal intervals require all 365 planned pairs and source checks; the fixed analysis cutoff excludes later-arriving scores.

A separate option-distribution study uses stochastic volatility inspired (SVI) option smiles to define probability forecasts for later settlement outcomes. It has zero settled research forecasts. Its original protocol and baseline locks remain unchanged. A versioned amendment sets the expiry cutoff at 20 October 2027, 08:00 UTC and the final decision at 22 October 2027, 08:00 UTC. Interim reporting and command-line inference remain descriptive, with terminal data availability fixed.

A Infrastructure and reproduction

Immutable raw capture segments, code hashes and input hashes accompany the forecasts. Recoverable delivery-pagination failures trigger retries in fresh segments; conflicting immutable prices remain fatal. Cleaning is cached by input, protocol, code and runtime fingerprints. Reports render outside the forecast loop. Published operational snapshots contain counts, rejection reasons, local hashes and heartbeat age.

These operational signals have distinct limits: local forecast hash chains lack an independent public timestamp, and a fresh heartbeat alone leaves process health unverified. Continuous collection requires an awake online host. The short validation recordings demonstrate bounded operation rather than sustained collection.

The numerical rebuild uses Python 3.12 with `requirements-research.txt` and runs:

```
python scripts/build_research.py
```

The command reads the checked-in public recordings without market requests and retains the full report, matched forecasts, coefficients, training cutoffs, diagnostics, simulation rates and plan hashes. The manuscript compiles separately from its checked LaTeX source, bibliography and figure PDF.

References.

Corsi (2009), A Simple Approximate Long-Memory Model of Realized Volatility, doi:10.1093/jjfinec/nbp001.

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Deribit, DVOL methodology, official methodology.